



Barker College

**2001
TRIAL
HIGHER SCHOOL
CERTIFICATE**

Mathematics Extension 2

AM WEDNESDAY 8 AUGUST

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using blue or black pen
- Make sure your Barker Student Number is on ALL pages
- Board-approved calculators may be used
- A table of standard integrals is provided on page 10.
- ALL necessary working should be shown in every question

Total marks (120)

- Attempt Questions 1 – 8
- All questions are of equal value

Total marks (120)

Attempt Questions 1 – 8

ALL questions are of equal value

Answer each question on a SEPARATE sheet of paper

Marks

Question 1 [15 marks] [START A NEW PAGE]

(a) Find

$$\int \frac{dx}{x^2 - 16x + 80}$$

2

(b) Evaluate

(i) $\int_0^{\frac{\pi}{2}} \frac{1}{1 + \sin \theta} d\theta$

3

(ii) $\int_0^2 \frac{8 dx}{(x + 2)(x^2 + 4)}$

4

(iii) $\int_0^{\pi} e^x \cos x dx$

3

(c) Find $\int \frac{2x}{\sqrt{4x - x^2}} dx$

You may wish to use the substitution of $u = x - 2$.

3

Question 2 [15 marks] [START A NEW PAGE]

(a) (i) Find all real numbers x and y such that $(x + iy)^2 = -3 + 4i$. 2

(ii) Hence, solve the equation $z^2 - 3z + (3 - i) = 0$. 1

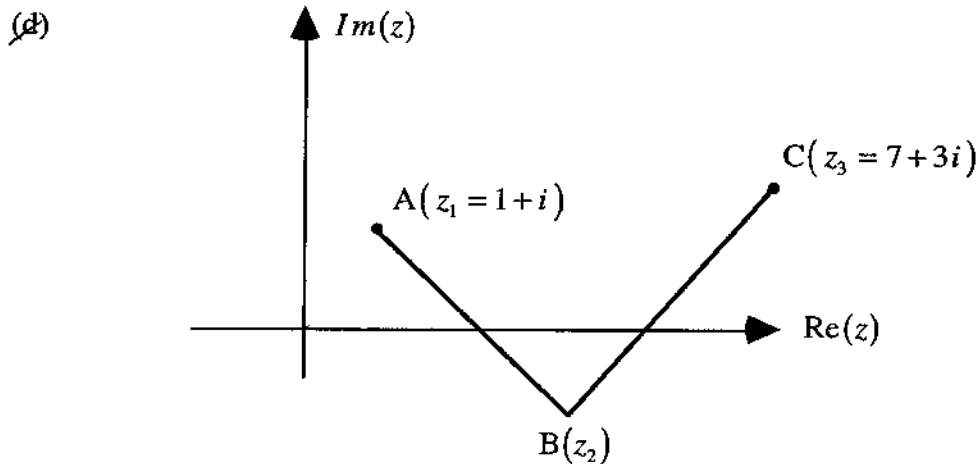
(b) (i) Express $\sqrt{3} + i$ and $\sqrt{3} - i$ in modulus-argument form. 2

(ii) Hence, simplify $(\sqrt{3} + i)^{15} + (\sqrt{3} - i)^{15}$ 1

(c) Sketch the locus specified by

(i) $|z| \leq |z - 2|$ and $-\frac{\pi}{4} \leq \arg z \leq \frac{\pi}{4}$ 3

(ii) $\operatorname{Re}\left(z - \frac{1}{z}\right) = 0$ (State the equation(s) of the locus). 4



The points A and C represent the complex numbers

$$z_1 = 1 + i \text{ and } z_3 = 7 + 3i$$

Find the complex number z_2 represented by B such that $\triangle ABC$ is isosceles and right angled at B.

2

Question 3 [15 marks] [START A NEW PAGE]

(a) If $f(x) = (x - 1)(x - 3)$ then sketch

(i) $y = \frac{1}{f(x)}$ 2

(ii) $y = f(|x|)$ 2

(iii) $|y| = f(x)$ 2

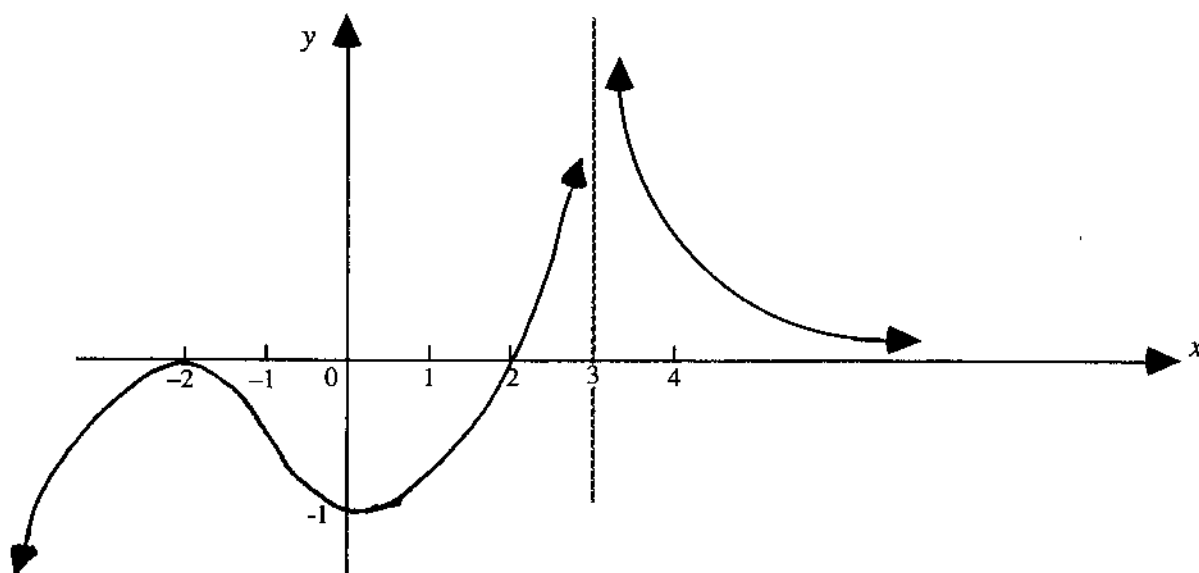
(b) (i) Find the stationary points and the asymptotes of the function

$$y = \frac{(x + 1)^4}{x^4 + 1}$$
 2

(ii) Sketch this function labelling all essential features. 1

(iii) Use the graph to find the set of values of k for which $(x + 1)^4 = k(x^4 + 1)$ has two distinct real roots. 2

(c) Given the graph of $y = f'(x)$ below, sketch the graph of $y = f(x)$.
 $y = f'(x)$ is the derivative of $y = f(x)$. 4



Question 4 [15 marks] [START A NEW PAGE]

(a) An ellipse has the equation $\frac{x^2}{16} + \frac{y^2}{9} = 1$

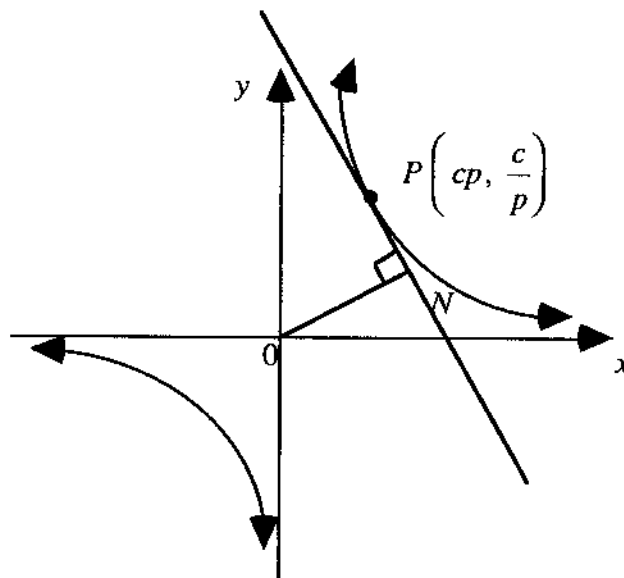
(i) Sketch the ellipse showing the foci S and S' and the directrices. 4

(ii) Prove that the tangent at the point $P(4\cos\theta, 3\sin\theta)$ to the ellipse has the equation $\frac{x\cos\theta}{4} + \frac{y\sin\theta}{3} = 1$ 3

(iii) The ellipse meets the y -axis at B and B' . The tangents at B and B' meet the tangent at P at the points Q and Q' .
Prove that $BQ \cdot B'Q' = 16$ 3

(b) The line through O perpendicular to the tangent at $P\left(cp, \frac{c}{p}\right)$ on the rectangular hyperbola $xy = c^2$ meets the tangent at N .

Find the coordinates of N and show that as p varies, the locus of N is $(x^2 + y^2)^2 = 4c^2xy$. 5



Question 5 [15 marks] [START A NEW PAGE]

(a) A solid has as its base the ellipse $\frac{x^2}{36} + \frac{y^2}{16} = 1$

If each section perpendicular to the major axis is an equilateral triangle, show that the volume of the solid is $128\sqrt{3}$ units³.

4

(b) The region $(x - 2R)^2 + y^2 \leq R^2$ is rotated about the y-axis forming a solid of revolution called a torus.

By summing volumes of cylindrical shells, show that the volume of the torus is $4\pi^2 R^3$ units³.

6

(c) The angles of elevation of the top of a tower P measured from three points A, B, C are α , β , γ respectively. A, B, C are in a straight line such that $AB = BC = a$, but the line AC does not pass through S, the base of the tower.

(i) If $\angle ABS = \theta$, show that

$$(CS)^2 = a^2 + h^2 \cot^2 \beta + 2ah \cot \beta \cos \theta$$

2

(ii) Prove that the height of the tower is

$$\frac{a\sqrt{2}}{\{\cot^2 \alpha + \cot^2 \gamma - 2\cot^2 \beta\}^{\frac{1}{2}}}$$

3

Question 6 [15 marks] [START A NEW PAGE]

- (a) Given that a, b and c are the roots of the equation $x^3 + qx + r = 0$, find the cubic equation in y , in terms of q and r , whose roots are $(b + c - 2a), (c + a - 2b)$ and $(a + b - 2c)$

3

- (b) Using $\tan 3\theta = \tan(2\theta + \theta)$, show that

$$\tan 3\theta = \frac{3 \tan \theta - \tan^3 \theta}{1 - 3 \tan^2 \theta}$$

2

- (ii) Find the value of x for which $3 \tan^{-1} x = \frac{\pi}{2} - \tan^{-1}(3x)$ where $\tan^{-1} x$ and $\tan^{-1}(3x)$ both lie between 0 and $\frac{\pi}{2}$

4

- (c) Using mathematical induction, prove that

$$\sum_{r=0}^n \frac{1}{(r+1)(r+2)(r+3)} = \frac{(n+1)(n+4)}{4(n+2)(n+3)}$$

4

- (iii) Hence, find $\lim_{n \rightarrow \infty} \sum_{r=0}^n \frac{1}{(r+1)(r+2)(r+3)}$

2

Question 7 [15 marks] [START A NEW PAGE]

(a) ~~(i)~~ Given that $\int_0^\pi \sin^n \theta d\theta = I_n$, prove that

$$nI_n = (n - 1)I_{n-2}$$

3

~~(ii)~~ Hence, evaluate I_8

1

~~(iii)~~ Use the result $\int_0^a f(x)dx = \int_0^a f(a - x)dx$ to show that

$$\int_0^\pi x \sin^n x dx = \left(\frac{\pi}{2}\right) I_n$$

2

(b) ~~(i)~~ Use De Moivre's Theorem to prove that, if $2\cos\theta = x + \frac{1}{x}$,
then $2\cos n\theta = x^n + \frac{1}{x^n}$

1

~~(ii)~~ Hence, or otherwise, solve the equation

$$5x^4 - 11x^3 + 16x^2 - 11x + 5 = 0$$

4

(c) At the ends of three successive seconds, the distances of a point moving with Simple Harmonic Motion from its mean position, measured in the same direction, are 1, 5 and 5 metres.

Show that the period of the complete oscillation is $\frac{2\pi}{\cos^{-1}\left(\frac{3}{5}\right)}$ seconds.

4

Question 8 [15 marks] [START A NEW PAGE]

- (a) A ball thrown from a point P with velocity V , at an inclination α to the horizontal reaches a point Q after t seconds.
 Show that if PQ is inclined at θ to the horizontal, (where $\alpha > \theta$), then the direction of motion of the ball, when at Q , is inclined to the horizontal at an acute angle of $\tan^{-1}[2 \tan \theta - \tan \alpha]$.

You may use the result without proof

$$x = V \cos \alpha \times t$$

$$y = V \sin \alpha \times t - \frac{1}{2} g t^2$$

4

- (b) (i) A gun fires shells with muzzle velocity V . Ignoring air resistance, show that the range on a horizontal plane is $\frac{V^2 \sin 2\theta}{g}$ where θ is the angle of elevation of the gun and g is the acceleration due to gravity.

2

- (ii) The gun and the target lie on the same horizontal plane. The gun fires, in the correct vertical plane, at the target using an angle of elevation α and the shell falls short by a distance p . When the angle of elevation is changed to β , the shell overshoots the target by a distance q .

$$\text{Show that } \sin 2\theta = \frac{p \sin 2\beta + q \sin 2\alpha}{p + q}$$

4

- (c) Fred has three uniform tetrahedra (triangular pyramids). Each of these tetrahedra has one face black, one face white, one face red and one face green.
 When tossed onto a table, three faces of each tetrahedron can be seen.
 If the probability of any coloured face not being seen is equally likely, what is the probability that

- (i) no black face can be seen?

1

- (ii) exactly 2 black faces can be seen?

1

- (iii) at least 2 red faces can be seen?

1

- (iv) 3 white faces and only 1 green face can be seen?

2

END OF PAPER

Question 1

$$(a) \int \frac{dx}{x^2 - 16x + 80}$$

$$= \int \frac{dx}{(x-8)^2 + 16}$$

$$= \frac{1}{4} \tan^{-1} \left(\frac{x-8}{4} \right) + C$$

$$(b) \text{ii) } \int_0^{\frac{\pi}{2}} \frac{1}{1 + \sin \theta} d\theta$$

$$= \int_0^1 \frac{2}{(t+1)^2} dt \quad \text{let } \tan \frac{\theta}{2} = t$$

$$= -2 \left[\frac{1}{t+1} \right]_0^1 \quad \frac{dt}{d\theta} = \frac{1}{2}(1+t^2)$$

$$= 1$$

$$(ii) \int_0^2 \frac{8 dx}{(x+2)(x^2+4)}$$

$$\frac{1}{(x+2)(x^2+4)} = \frac{A}{x+2} + \frac{Bx+C}{x^2+4}$$

$$A = \frac{1}{8}, B = -\frac{1}{8}, C = \frac{1}{4}$$

$$I = \int_0^2 \left(\frac{1}{8(x+2)} + \frac{-\frac{1}{8}x + \frac{1}{4}}{x^2+4} \right) dx$$

$$= \left[\ln(x+2) \right]_0^2 - \frac{1}{16} \int_0^2 \frac{x}{x^2+4} dx + 2 \int_0^2 \frac{1}{x^2+4} dx$$

$$= \ln 2 - \frac{1}{32} \left[\ln(x^2+4) \right]_0^2 + \left[\tan^{-1} \left(\frac{x}{2} \right) \right]_0^2$$

$$= \frac{\pi}{4} + \frac{1}{32} \ln 2$$

$$(iii) I = \int_0^{\pi} e^x \cos x dx$$

$$= e^x \cos x + \int e^x \sin x dx$$

$$= e^x \cos x + [\sin x e^x - \int e^x \cos x dx]$$

$$= e^x \cos x + \sin x e^x - I$$

$$\therefore 2I = \left[e^x (\cos x + \sin x) \right]_0^{\pi}$$

$$I = \frac{1}{2} [-e^{\pi} - 1]$$

$$(c) \int \frac{2x}{\sqrt{4x-x^2}} dx \quad \text{let } u = x-2 \quad du = dx$$

$$= \int \frac{2(u+2)}{\sqrt{4(u+2)-(u+2)^2}} du$$

$$= 2 \int \frac{u+2}{\sqrt{4-u^2}} du$$

$$= 2 \int \frac{u}{\sqrt{4-u^2}} du + 4 \int \frac{1}{\sqrt{4-u^2}} du$$

$$= -2\sqrt{4-u^2} + 4 \sin^{-1} \left(\frac{u}{2} \right) + C$$

$$= -2\sqrt{4x-x^2} + 4 \sin^{-1} \left(\frac{x-2}{2} \right) + C$$

Question 2

$$(a) (x+iy)^2 = -3+4i$$

$$x^2 - y^2 + 2xyi = -3+4i$$

$$\begin{cases} x^2 - y^2 = -3 \\ xy = 2 \end{cases}$$

$$x^2 - y^2 = -3$$

solving simultaneous.

$$\begin{cases} x=2, y=1 \\ x=-1, y=-2 \end{cases}$$

$$(b) z^2 - 3z + (3-i) = 0$$

$$z = \frac{3 \pm \sqrt{9-4(3-i)}}{2}$$

$$= \frac{3 \pm \sqrt{-3+4i}}{2}$$

$$= 2+i \text{ or } 1-i$$

Question 2

$$(b) \text{ii) } \sqrt{3}+i = 2 \left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6} \right)$$

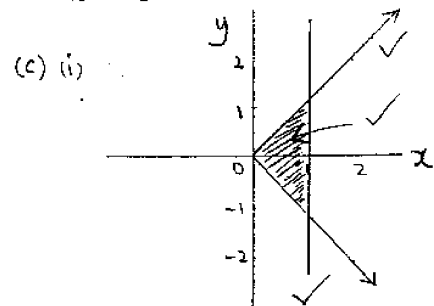
$$\sqrt{3}-i = 2 \left(\cos \frac{\pi}{6} - i \sin \frac{\pi}{6} \right)$$

$$\text{iii) } (\sqrt{3}+i)^{15} + (\sqrt{3}-i)^{15}$$

$$= 2^{15} \left(\cos \frac{15\pi}{6} + i \sin \frac{15\pi}{6} \right) + 2^{15} \left(\cos \frac{15\pi}{6} - i \sin \frac{15\pi}{6} \right)$$

$$= 2 \times 2^{15} \cos \frac{15\pi}{6}$$

$$= 0$$



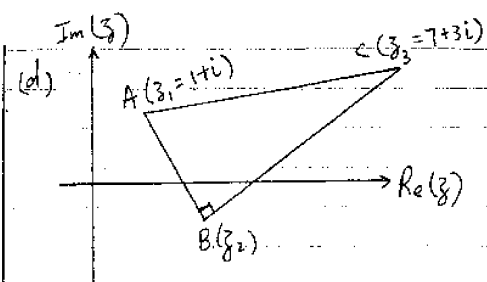
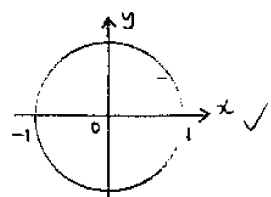
$$(ii) \operatorname{Re} \left(z - \frac{1}{z} \right) = 0$$

$$\operatorname{Re} \left(x+iy - \frac{x-iy}{x^2+y^2} \right) = 0$$

$$x - \frac{x}{x^2+y^2} = 0$$

$$x^2 + y^2 = 1$$

or $x=0$



$$z_2 = x+iy$$

$$i \vec{BC} = \vec{BA}$$

$$i[(7-x) + (3-y)i] = (1-x) + i(1-y)$$

$$-(3-y) + i(7-x) = (1-x) + i(1-y)$$

Equating real part,

$$\begin{cases} -3+y = 1-x \\ x+y = 4 \end{cases} \quad (1)$$

Equating imaginary part,

$$\begin{cases} 7-x = 1-y \\ -x+y = -6 \end{cases} \quad (2)$$

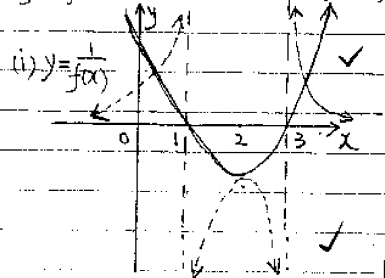
solving (1) + (2) simult.

$$x=5, y=-1$$

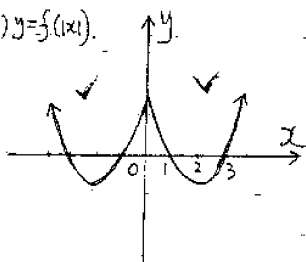
$$z_2 = 5-i$$

Question 3

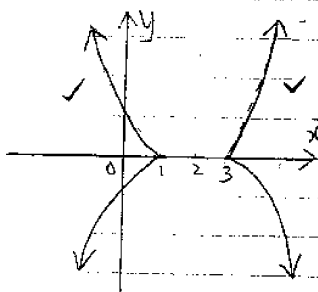
(a) $f(x) = (x-1)(x-3)$



(ii) $y = f(x)$



(iii) $|y| = f(x)$



(b) (i) $y = \frac{(x+1)^4}{x^4+1}$

$$\frac{dy}{dx} = \frac{(x^4+1)^4(x+1)^3 - (x+1)^4 \cdot 4x^3}{(x^4+1)^2}$$

$$= \frac{4(x+1)^3(1-x^3)}{(x^4+1)^2}$$

$y' = 0 \Rightarrow x = -1, 1$

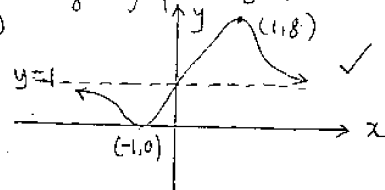
Stationary points are

$(-1, 0), (1, 8)$

As $x \rightarrow \infty, y \rightarrow 1$

Horizontal asymptote is $y = 1$

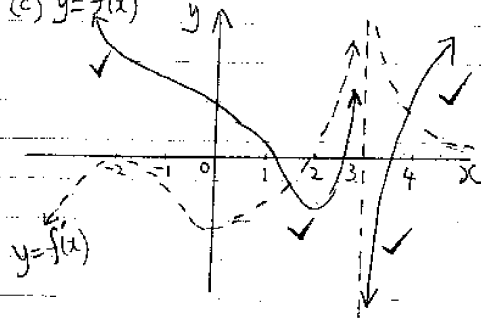
(ii)



(iii) $(x+1)^4 = K(x^4+1)$ has two distinct real roots

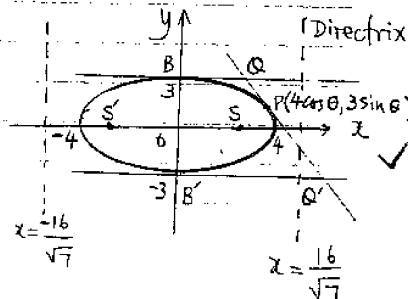
$\therefore$ the line $y = K$ must meet the curve at two points
 $0 < K < 1$ and $1 < K < 8$

(c) $y = f(x)$



Question 4

(a) (i) $\frac{x^2}{16} + \frac{y^2}{9} = 1$



$b^2 = a^2(1-e^2)$

$9 = 16(1-e^2)$

$e = \frac{\sqrt{7}}{4}$

$S(ae, 0) = (\sqrt{7}, 0)$

$S'(-\sqrt{7}, 0)$

Directrix $x = \pm \frac{a}{e}$
 $= \pm \frac{16}{\sqrt{7}}$

(ii) $\frac{dy}{dx} = \frac{-9x}{16y}$
 $= -\frac{3 \cot \theta}{4}$

Equation of the tangent at P

$y - 3 \sin \theta = \frac{-3 \cot \theta}{4} (x - 4 \cos \theta)$

$\frac{y \sin \theta}{3} + \frac{x \cos \theta}{4} = 1$

(iii) At Q, $y = 3$

$\therefore \frac{3 \sin \theta}{3} + \frac{x \cos \theta}{4} = 1$

$\therefore x_Q = \frac{4(1 - \sin \theta)}{\cos \theta}$

At Q', $y = -3$

$\therefore \frac{-3 \sin \theta}{3} + \frac{x \cos \theta}{4} = 1$

$x_{Q'} = \frac{4(1 + \sin \theta)}{\cos \theta}$

BQ x B'Q'

$= \frac{4(1 - \sin \theta)}{\cos \theta} \times \frac{4(1 + \sin \theta)}{\cos \theta} = 16$

(b) $y = \frac{c^2}{x}$
 $y' = -\frac{c^2}{x^2}$

Eqn of tangent at P,

$y - \frac{c}{p} = -\frac{1}{p^2} (x - cp)$

$x + p^2 y = 2cp$ (1)

Equation of ON is

$y = p^2 x$ (2)

Solving (1), (2) Simult to find coord N

$x_N = \frac{2cp}{1+p^4}$

$y_N = \frac{2cp^3}{1+p^4}$

using $x(1+p^4) = 2cp$ and $\frac{y}{x} = p^2$,

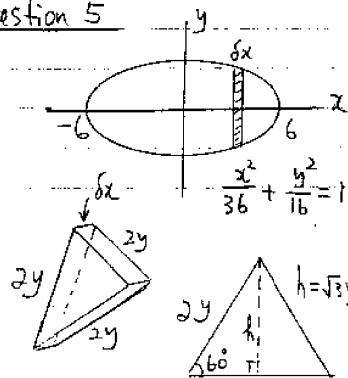
$x^2(1+p^4)^2 = (2cp)^2$

$x^2(1+\frac{y^2}{x^2})^2 = 4c^2(\frac{y}{x})$

$(x^2+y^2)^2 = 4c^2xy$

Question 5

(a)



cross-sectional area of $\Delta = \frac{1}{2} \sqrt{3} y^2$

$$\delta V = A \delta x$$

$$= \sqrt{3} y^2 \delta x \quad \checkmark$$

Vol of the solid

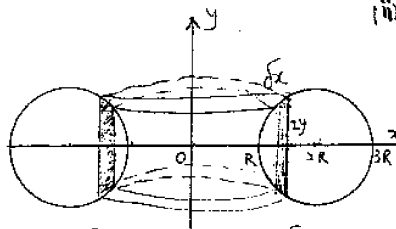
$$= \int \sqrt{3} y^2 dx$$

$$= \sqrt{3} \int_{-6}^6 16 \left(1 - \frac{x^2}{36}\right) dx \quad \checkmark$$

$$= 16\sqrt{3} \left[x - \frac{x^3}{108} \right]_{-6}^6 \quad \checkmark$$

$$= 128\sqrt{3} \text{ units}^3 \quad \checkmark$$

(b)



$$\delta V = 2\pi x \cdot 2y \delta x$$

$$= 4\pi x y \delta x \quad \checkmark$$

$$\text{Volume} = \int_R^{3R} 4\pi x \sqrt{R^2 - (x-2R)^2} dx \quad \checkmark$$

$$\text{let } x-2R = R \sin \theta$$

$$dx = R \cos \theta d\theta$$

volume of the torus

$$= 4\pi \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sqrt{R^2 - R^2 \sin^2 \theta} \cdot (2R + R \sin \theta) R \cos \theta d\theta \quad \checkmark$$

$$= 4\pi R^3 \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (2 \cos^2 \theta + \cos^2 \theta \sin \theta) d\theta$$

$$= 4\pi R^3 \left[\left(\theta + \sin 2\theta \right) - \frac{1}{3} \cos^3 \theta \right]_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \quad \checkmark \checkmark$$

$$= 4\pi R^3 \quad \checkmark$$

(c)

$$\left. \begin{aligned} AS &= h \cot \alpha \\ BS &= h \cot \beta \\ CS &= h \cot \gamma \end{aligned} \right\} \quad \checkmark$$

$$\angle SBC = 180^\circ - \theta$$

$$\angle ABS = \theta$$

$$\text{In } \triangle BSC, (SC)^2 = a^2 + h^2 \cot^2 \beta - 2ah \cot \beta \cos(180^\circ - \theta)$$

$$h^2 \cot^2 \gamma = a^2 + h^2 \cot^2 \beta + 2ah \cot \beta \cos \theta \quad \text{--- (1)} \quad \checkmark$$

$$\text{In } \triangle ABS, h^2 \cot^2 \alpha = a^2 + h^2 \cot^2 \beta - 2ah \cot \beta \cos \theta \quad \text{--- (2)} \quad \checkmark$$

$$\text{Adding, (1) + (2)}$$

$$h^2 (\cot^2 \alpha + \cot^2 \gamma) = 2a^2 + 2h^2 \cot^2 \beta \quad \checkmark$$

$$R^2 (\cot^2 \alpha + \cot^2 \gamma - 2 \cot^2 \beta) = 2a^2$$

$$h = \frac{a\sqrt{2}}{[\cot^2 \alpha + \cot^2 \gamma - 2 \cot^2 \beta]^{\frac{1}{2}}} \quad \checkmark$$

Question 6

(a) a, b, c are roots of $x^3 + qx + r = 0$

$$\text{let } y = b+c-a$$

$$= (b+c+a) - 3a$$

$$= 0 - 3a$$

$$a = -\frac{y}{3} \quad \checkmark$$

$$\left(-\frac{y}{3}\right)^3 + q\left(-\frac{y}{3}\right) + r = 0 \quad \checkmark$$

$$y^3 + 9qy - 27r = 0 \quad \checkmark$$

(b) (i) $\tan 3\theta = \tan(2\theta + \theta)$

$$\text{RHS} = \frac{\tan 2\theta + \tan \theta}{1 - \tan 2\theta \tan \theta} \quad \checkmark$$

$$= \frac{\frac{2 \tan \theta}{1 - \tan^2 \theta} + \tan \theta}{1 - \frac{2 \tan \theta}{1 - \tan^2 \theta} \tan \theta}$$

$$= \frac{3 \tan \theta - \tan^3 \theta}{1 - 3 \tan^2 \theta} \quad \checkmark$$

(ii) let $\theta = \tan^{-1} x$

$$\therefore \tan(3 \tan^{-1} x) = \frac{3x - x^3}{1 - 3x^2}$$

$$\text{using } 3 \tan^{-1} x = \frac{\pi}{2} - \tan^{-1}(3x) \quad \checkmark$$

$$\tan(3 \tan^{-1} x) = \tan\left(\frac{\pi}{2} - \tan^{-1}(3x)\right) \quad \checkmark$$

$$= \cot(\tan^{-1}(3x))$$

$$= \frac{1}{\tan[\tan^{-1}(3x)]} \quad \checkmark$$

$$\frac{3x - x^3}{1 - 3x^2} = \frac{1}{3x}$$

$$3x^4 - 12x^2 + 1 = 0 \quad \checkmark$$

$$x^2 = \frac{12 \pm \sqrt{144 - 12}}{6}$$

$$\therefore x = 0.292 \text{ only} \quad \checkmark$$

$$x = 1.979 \text{ is not the solution.}$$

(c) (i) $\frac{1}{1 \cdot 2 \cdot 3} + \frac{1}{2 \cdot 3 \cdot 4} + \dots$

$$+ \frac{1}{(n+1)(n+2)(n+3)}$$

$$= \frac{(n+1)(n+4)}{4(n+2)(n+3)}$$

When $n=0$, LHS = $\frac{1}{6}$, RHS = $\frac{1}{6} \quad \checkmark$

Assume that it is true for $n=k$

$$\frac{1}{1 \cdot 2 \cdot 3} + \frac{1}{2 \cdot 3 \cdot 4} + \dots + \frac{1}{(k+1)(k+2)(k+3)}$$

$$= \frac{(k+1)(k+4)}{4(k+2)(k+3)} \quad \checkmark$$

$$\text{RTP: } S_{k+1} = S_k + T_{k+1}$$

$$= \frac{(k+2)(k+5)}{4(k+3)(k+4)}$$

$$S_{k+1} = \frac{1}{(k+2)(k+3)(k+4)} + \frac{(k+1)(k+4)}{4(k+2)(k+3)}$$

$$= \frac{4 + (k+1)(k+4)(k+4)}{4(k+3)(k+4)(k+2)}$$

$$= \frac{(k+1)(k^2 + 8k + 16) + 4}{4(k+2)(k+3)(k+4)}$$

$$= \frac{(k+2)^2(k+5)}{4(k+2)(k+3)(k+4)}$$

$$= \frac{(k+2)(k+5)}{4(k+3)(k+4)} \quad \checkmark$$

Since it is true for $n=0$,

it is proven true for $n=k+1$.

$\therefore$ it is true for $n=0+1=1$

true for $n=1+1=2$

$\therefore$ it is true for all $n \geq 0 \quad \checkmark$

$$(ii) \lim_{n \rightarrow \infty} \sum_{r=0}^n \frac{1}{(n+1)(r+2)(r+3)}$$

$$= \frac{(n+1)(n+4)}{4(n+2)(n+3)}$$

$$= \frac{\left(\frac{n}{n} + \frac{1}{n}\right)\left(\frac{n}{n} + \frac{4}{n}\right)}{4\left(\frac{n}{n} + \frac{2}{n}\right)\left(\frac{n}{n} + \frac{3}{n}\right)} \quad \checkmark$$

$$= \frac{1 \times 1}{4 \times 1 \times 1} = \frac{1}{4} \quad \checkmark$$

Question 7

(a) (i) $I_n = \int_0^\pi \sin^n \theta d\theta$
 $= \int_0^\pi \sin \theta \cdot \sin^{n-1} \theta d\theta$
 $= [-\cos \theta \cdot \sin^{n-1} \theta]_0^\pi - \int_0^\pi \cos \theta \cdot (n-1) \sin^{n-2} \theta d\theta$
 $= 0 + (n-1) \int_0^\pi \cos \theta \sin^{n-2} \theta d\theta$
 $= (n-1) \int_0^\pi (1 - \sin^2 \theta) \sin^{n-2} \theta d\theta$
 $= (n-1) \int_0^\pi \sin^{n-2} \theta d\theta - (n-1) \int_0^\pi \sin^n \theta d\theta$
 $= (n-1) I_{n-2} - (n-1) I_n$
 $n I_n = (n-1) I_{n-2}$ ✓

(i) $I_8 = \frac{7}{8} I_6$
 $= \frac{7}{8} \cdot \frac{5}{6} \cdot I_4$
 $= \frac{7}{8} \cdot \frac{5}{6} \cdot \frac{3}{4} \cdot I_2$
 $= \frac{7}{8} \cdot \frac{5}{6} \cdot \frac{3}{4} \cdot \frac{1}{2} \cdot I_0$
 $= \frac{7}{8} \cdot \frac{5}{6} \cdot \frac{3}{4} \cdot \frac{1}{2} \cdot \pi$
 $= \frac{35\pi}{128}$ ✓

(iii) $\int_0^\pi x \sin^n x dx = \int_0^\pi (\pi-x) \sin^n (\pi-x) dx$
 $= \int_0^\pi (\pi-x) \sin^n x dx$
 $= \pi \int_0^\pi \sin^n x dx - \int_0^\pi x \sin^n x dx$ ✓
 $2 \int_0^\pi x \sin^n x dx = \pi \int_0^\pi \sin^n x dx$
 $\therefore \int_0^\pi x \sin^n x dx = I_n \left(\frac{\pi}{2} \right)$ ✓

(b) $x = \cos \theta + i \sin \theta$

(i) $\frac{1}{x} = \cos \theta - i \sin \theta$

$\therefore x + \frac{1}{x} = \cos \theta + i \sin \theta + \cos \theta - i \sin \theta$
 $= 2 \cos \theta$

$x^n + \frac{1}{x^n} = \cos n\theta + i \sin n\theta + \cos n\theta - i \sin n\theta$
 $= 2 \cos n\theta$ ✓

(ii) $5x^4 - 11x^3 + 16x^2 - 11x + 5 = 0$
 $5\left(x^2 + \frac{1}{x^2}\right) - 11\left(x + \frac{1}{x}\right) + 16 = 0$ ✓

$5 \cdot 2 \cos^2 \theta - 11 \cdot 2 \cos \theta + 16 = 0$
 $10 \cos^2 \theta - 11 \cos \theta + 3 = 0$ ✓
 $(5 \cos \theta - 3)(2 \cos \theta - 1) = 0$

$\cos \theta = \frac{3}{5}$ or $\cos \theta = \frac{1}{2}$
 $x = \frac{3}{5} + \frac{4}{5}i$, $x = \frac{3}{5} - \frac{4}{5}i$ ✓
 $x = \frac{1}{2} + \frac{\sqrt{3}}{2}i$, $x = \frac{1}{2} - \frac{\sqrt{3}}{2}i$ ✓

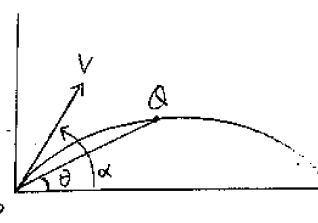
(c)

OB = 1 unit,
 OC = 5 units
 OB = a cos $\angle POB$
 $1 = a \cos \left(\omega + \frac{1}{2} \omega \right)$
 $= a \cos \frac{3\omega}{2}$ — (1) ✓
 OC = a cos $\angle OCA$
 $5 = a \cos \frac{\omega}{2}$ — (2) ✓

(a) $5 = \frac{\cos \frac{1}{2} \omega}{\cos \frac{3}{2} \omega}$
 $5 \cos 3\left(\frac{\omega}{2}\right) = \cos \frac{1}{2} \omega$
 $5 \left[4 \cos^3 \left(\frac{1}{2} \omega \right) - 3 \cos \frac{1}{2} \omega \right] = \cos \frac{1}{2} \omega$
 $\cos^2 \left(\frac{1}{2} \omega \right) = \frac{16}{20} \Rightarrow \cos^2 \left(\frac{\omega}{2} \right) = \frac{4}{5}$
 $\cos \omega = 2 \cos^2 \frac{1}{2} \omega - 1$
 $= \frac{3}{5}$ ✓
 Period $T = \frac{2\pi}{\omega} = \frac{2\pi}{\cos^{-1} \left(\frac{3}{5} \right)}$ ✓

Question 8

(a)



Let $\hat{x}$ and $\hat{y}$ be the horizontal and vertical components of velocity when the ball is at Q
 $\hat{x} = V \cos \alpha$, $\hat{y} = V \sin \alpha - gt$
 $\tan \theta = \frac{\hat{y}}{\hat{x}} = \frac{V \sin \alpha - \frac{1}{2} g t^2}{V \cos \alpha}$ ✓

$\therefore \tan \theta = \tan \alpha - \frac{gt}{2V \cos \alpha}$
 The direction of motion at Q is inclined to the horizontal at
 $\tan^{-1} \left(\frac{\hat{y}}{\hat{x}} \right) = \tan^{-1} \left[\tan \alpha - \frac{gt}{2V \cos \alpha} \right]$ ✓
 But $\frac{gt}{2V \cos \alpha} = 2 \tan \alpha - 2 \tan \theta$
 $\therefore \tan^{-1} \left(\frac{\hat{y}}{\hat{x}} \right) = \tan^{-1} [\tan \alpha - 2 \tan \alpha + 2 \tan \theta]$
 $= \tan^{-1} [2 \tan \theta - \tan \alpha]$ ✓

(b) (i) $x = V \cos \theta t$
 $y = V \sin \theta t - \frac{1}{2} g t^2$
 Time of flight $t \Rightarrow y = 0$
 $\Rightarrow t = \frac{2V \sin \theta}{g}$ ✓
 $\therefore \text{Range} = V \cos \theta \times \frac{2V \sin \theta}{g}$
 $R = \frac{V^2 \sin 2\theta}{g}$ ✓

(ii) $R - P = \frac{V^2 \sin 2\alpha}{g}$ — (1) ✓

$R + Q = \frac{V^2 \sin 2\beta}{g}$ — (2) ✓

$\frac{V^2 \sin 2\theta}{g} - P = \frac{V^2 \sin 2\alpha}{g}$

$\frac{V^2 \sin 2\theta}{g} + Q = \frac{V^2 \sin 2\beta}{g}$

$\frac{P}{Q} = \frac{\sin 2\theta - \sin 2\alpha}{\sin 2\beta - \sin 2\theta}$ ✓

$(P+Q) \sin 2\theta = P \sin 2\beta + Q \sin 2\alpha$
 $\sin 2\theta = \frac{P \sin 2\beta + Q \sin 2\alpha}{P+Q}$ ✓

(c) (i) $\left(\frac{1}{4} \right)^3 = \frac{1}{64}$ ✓

(ii) $\sum_2 \left(\frac{3}{4} \right)^2 \left(\frac{1}{4} \right)$
 $= \frac{27}{64}$ ✓

(iii) $\sum_2 \left(\frac{1}{4} \right)^2 \left(\frac{1}{4} \right) + \sum_3 \left(\frac{1}{4} \right)$
 $= \frac{27}{32}$ ✓

(iv) $\sum_2 \left(\frac{1}{4} \right) \left(\frac{1}{4} \right) \left(\frac{2}{4} \right)$
 $= \frac{3}{32}$ ✓

— END OF PAPER —